



Swiss Institute of
Bioinformatics

INTRODUCTION TO SEQUENCING-BASED
TRANSCRIPTOMICS DATA ANALYSIS

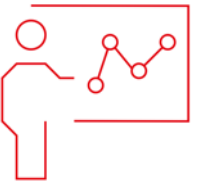
Introduction to spatial transcriptomics techniques

Joana Carlevaro Fita

June 3-5, 2026



Learning objectives

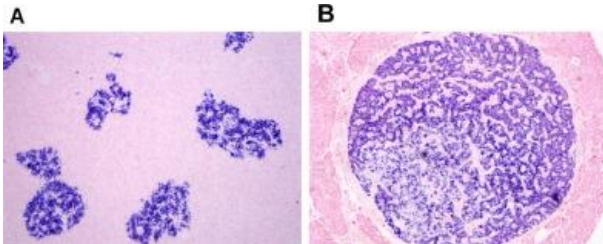
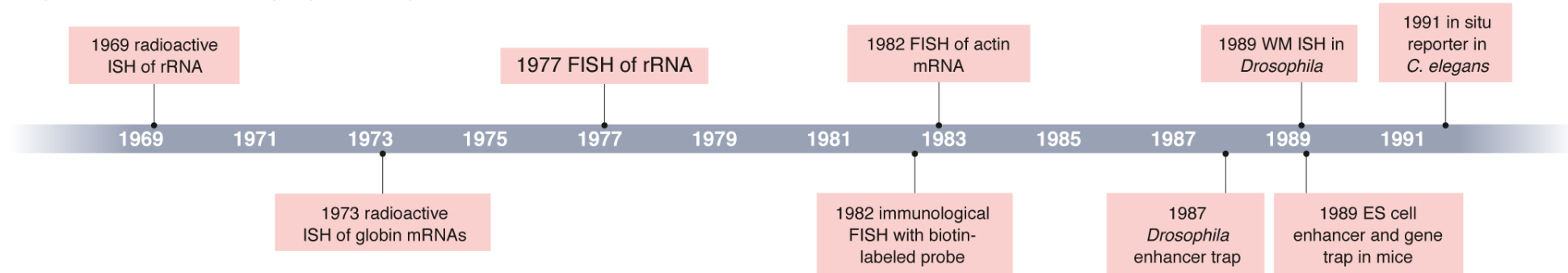


- ❖ Describe the general differences between imaging-based and sequencing-based ST methods
- ❖ Explain main features of 10X Visium and Xenium technologies
- ❖ Identify important criteria to choose the adequate technology

Introduction to spatial transcriptomics

History of spatial transcriptomics

a Major events in evolution of prequel techniques



Introduction to spatial transcriptomics

FISSEQ

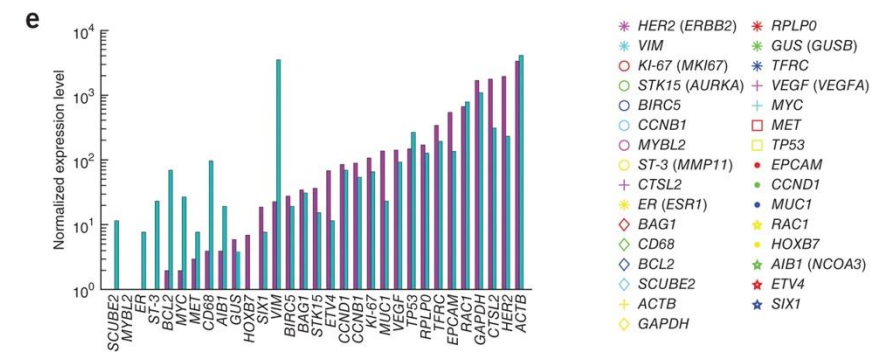
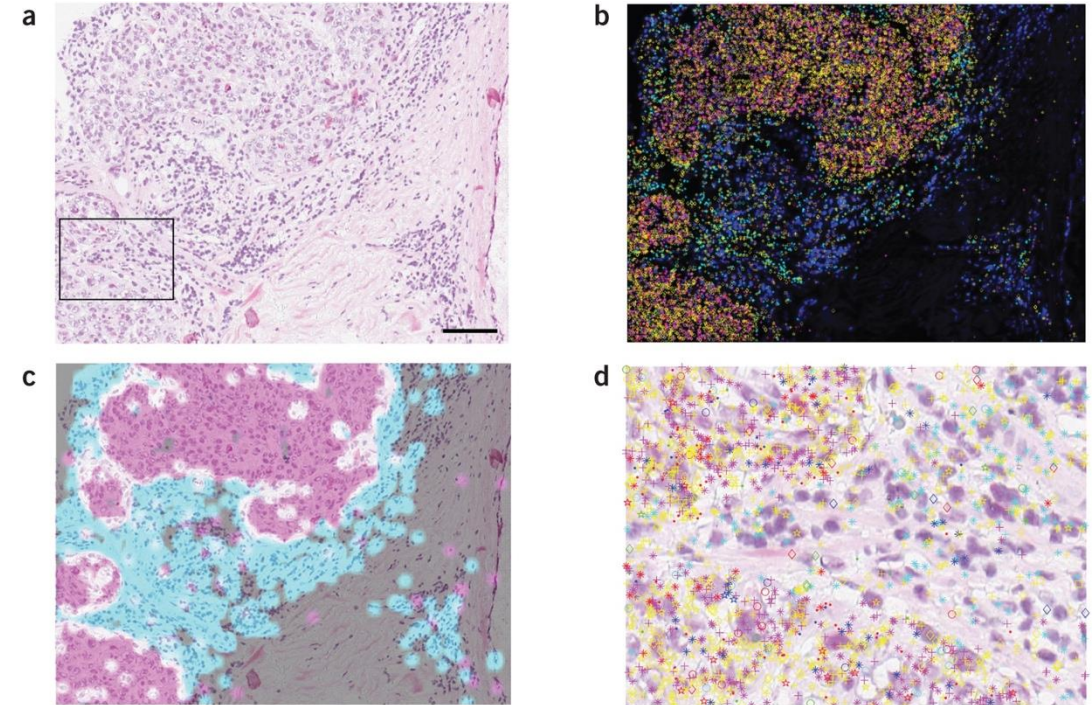
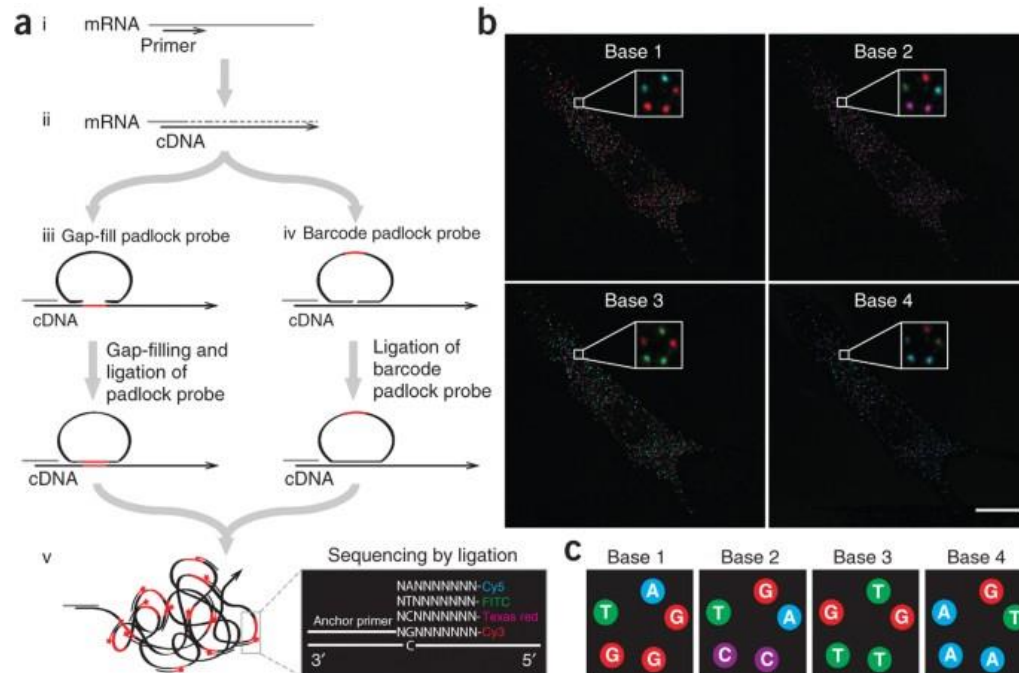
Brief Communication | Published: 14 July 2013

***In situ* sequencing for RNA analysis in preserved tissue and cells**

[Ronggin Ke](#), [Marco Mignardi](#), [Alexandra Pacureanu](#), [Jessica Svedlund](#), [Johan Botling](#), [Carolina Wählby](#)

✉ & [Mats Nilsson](#) ✉

[Nature Methods](#) **10**, 857–860 (2013) | [Cite this article](#)



Introduction to spatial transcriptomics

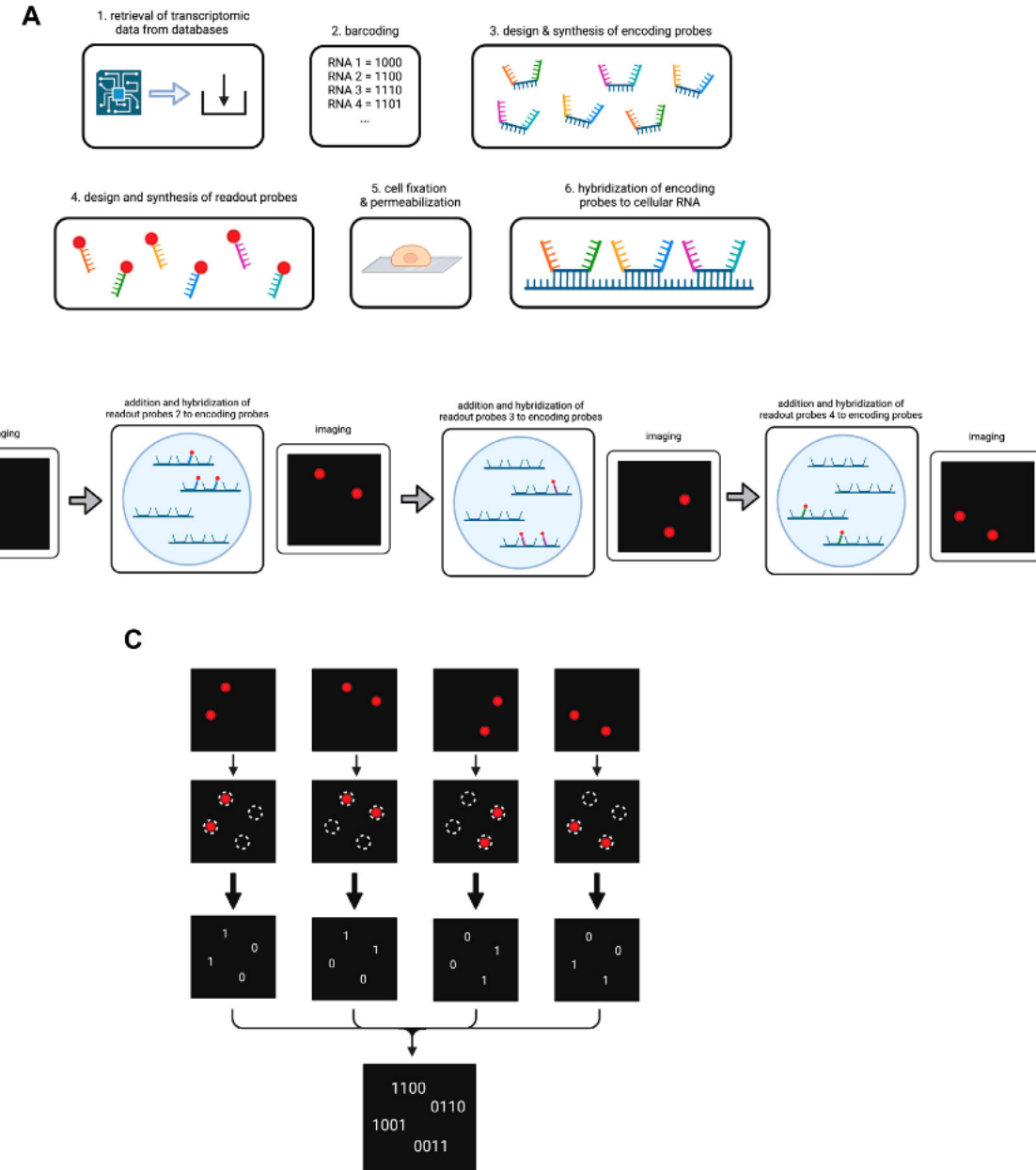
MERFISH

RESEARCH ARTICLE | SYSTEMS BIOLOGY | 



High-throughput single-cell gene-expression profiling with multiplexed error-robust fluorescence in situ hybridization

Jeffrey R. Moffitt , Junjie Hao, Guiping Wang, , and Xiaowei Zhuang   [Authors Info & Affiliations](#)



Introduction to spatial transcriptomics

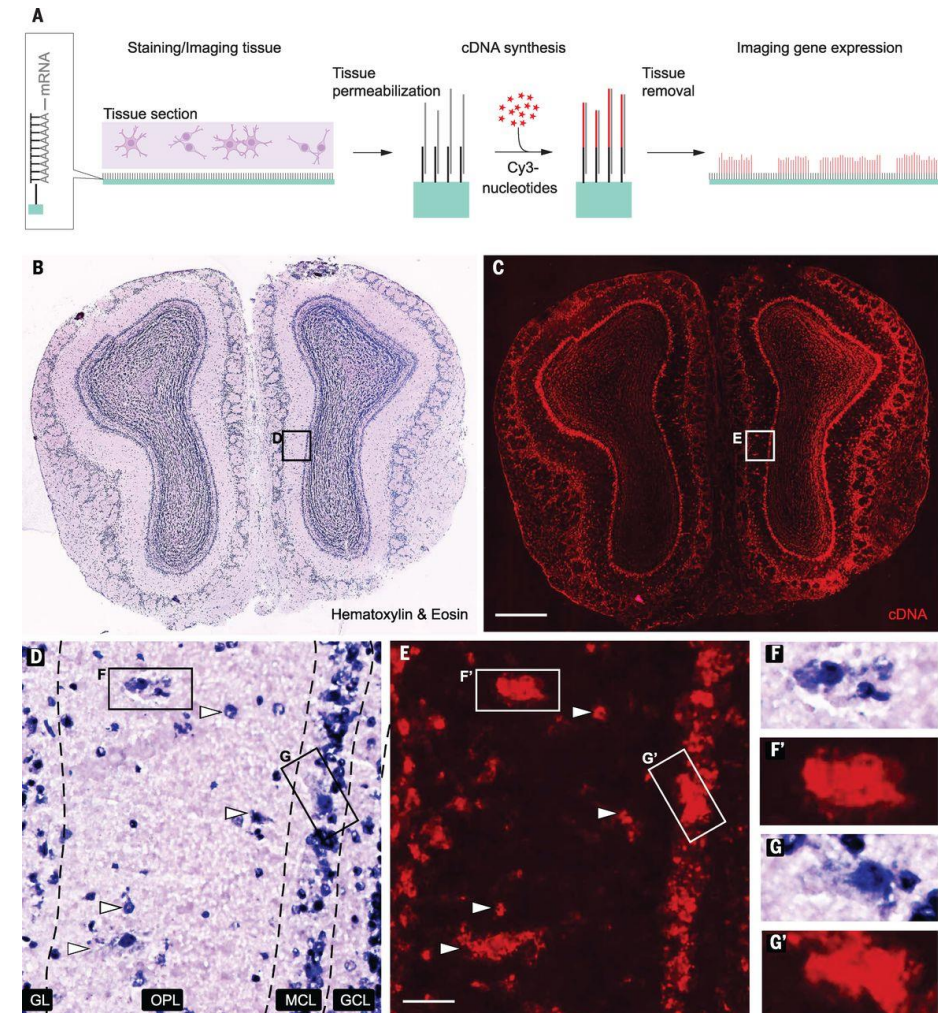
Spatial transcriptomics (ST)

Visualization and analysis of gene expression in tissue sections by spatial transcriptomics

PATRIK L. STÅHL, FREDRIK SALMÉN, SANJA VICKOVIC, ANNA LUNDMARK, JOSÉ FERNÁNDEZ NAVARRO, JENS MAGNUSSON, STEFANIA GIACOMELLO, MICHAELA ASP,

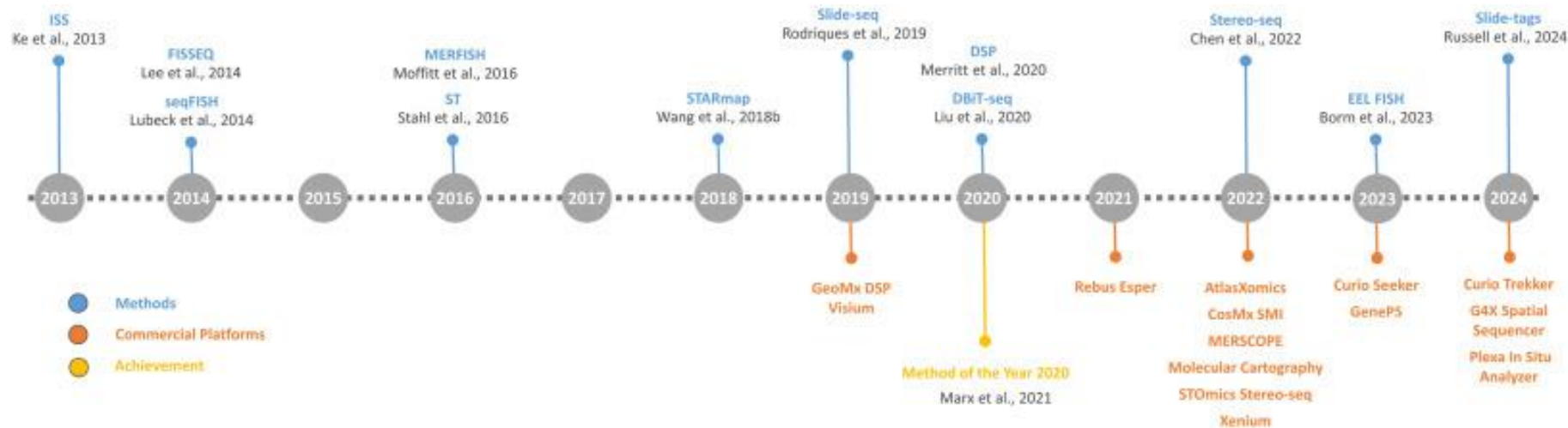
JAKUB O. WESTHOLM, [...] , AND JONAS FRISÉN [+11 authors](#) [Authors Info & Affiliations](#)

SCIENCE • 1 Jul 2016 • Vol 353, Issue 6294 • pp. 78-82 • DOI: [10.1126/science.aaf2403](https://doi.org/10.1126/science.aaf2403)

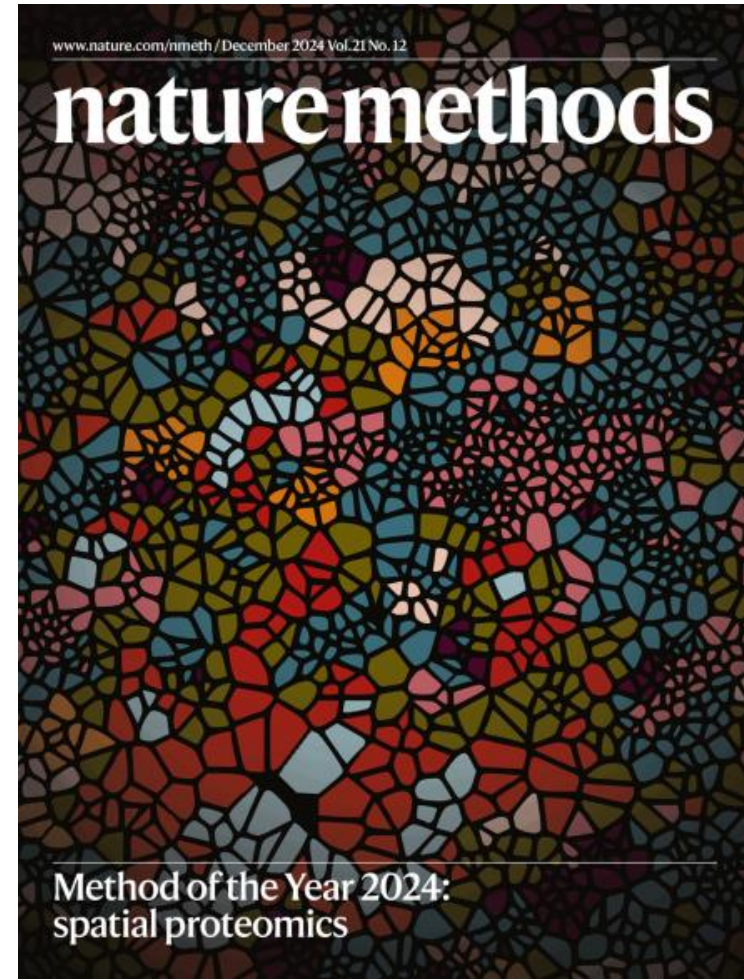
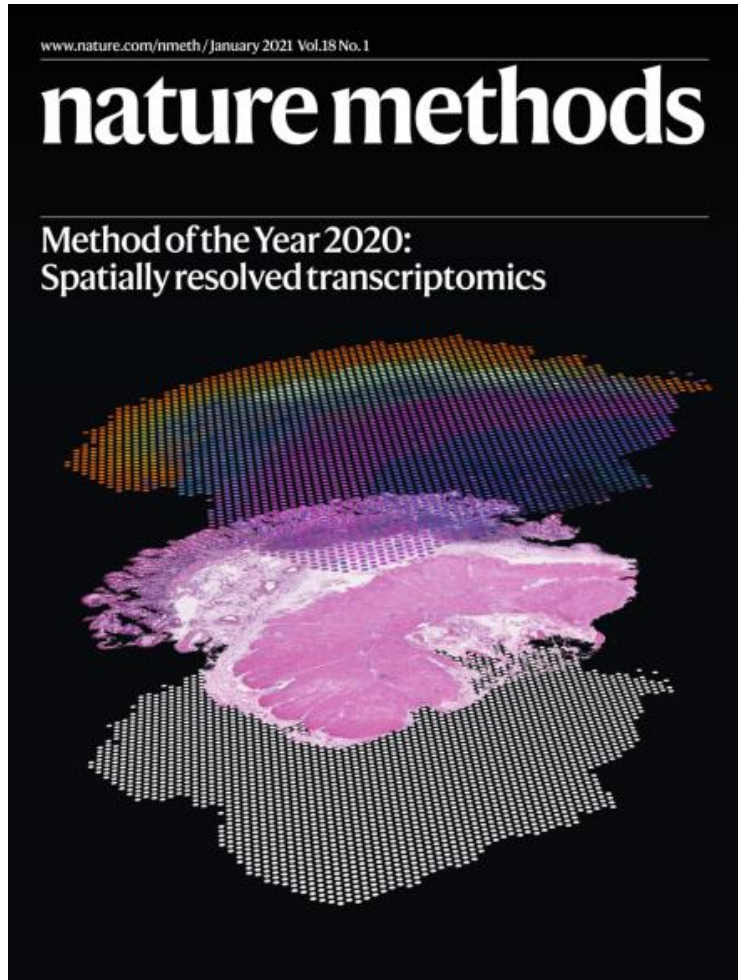


Introduction to spatial transcriptomics

Methods and commercialisation over the last years

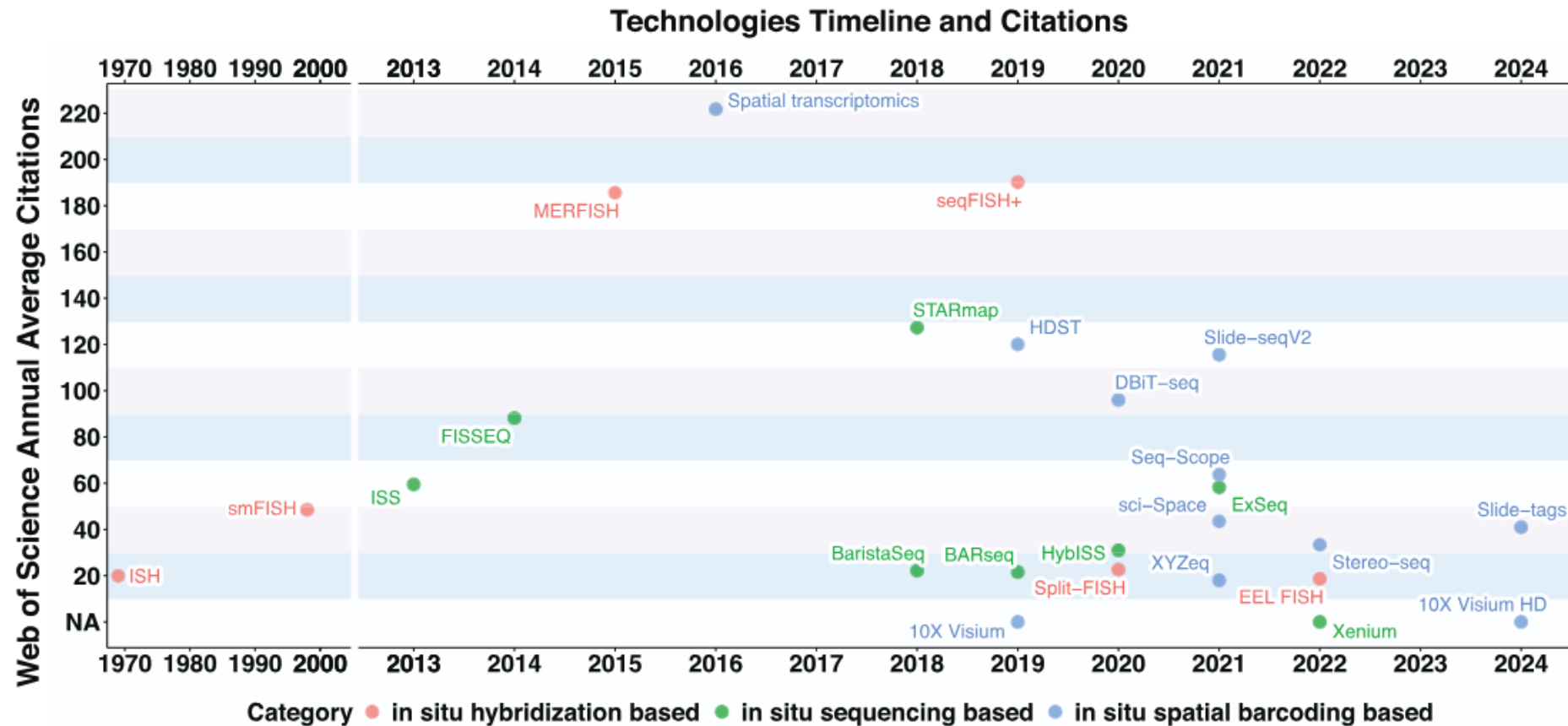


Spatial Omics

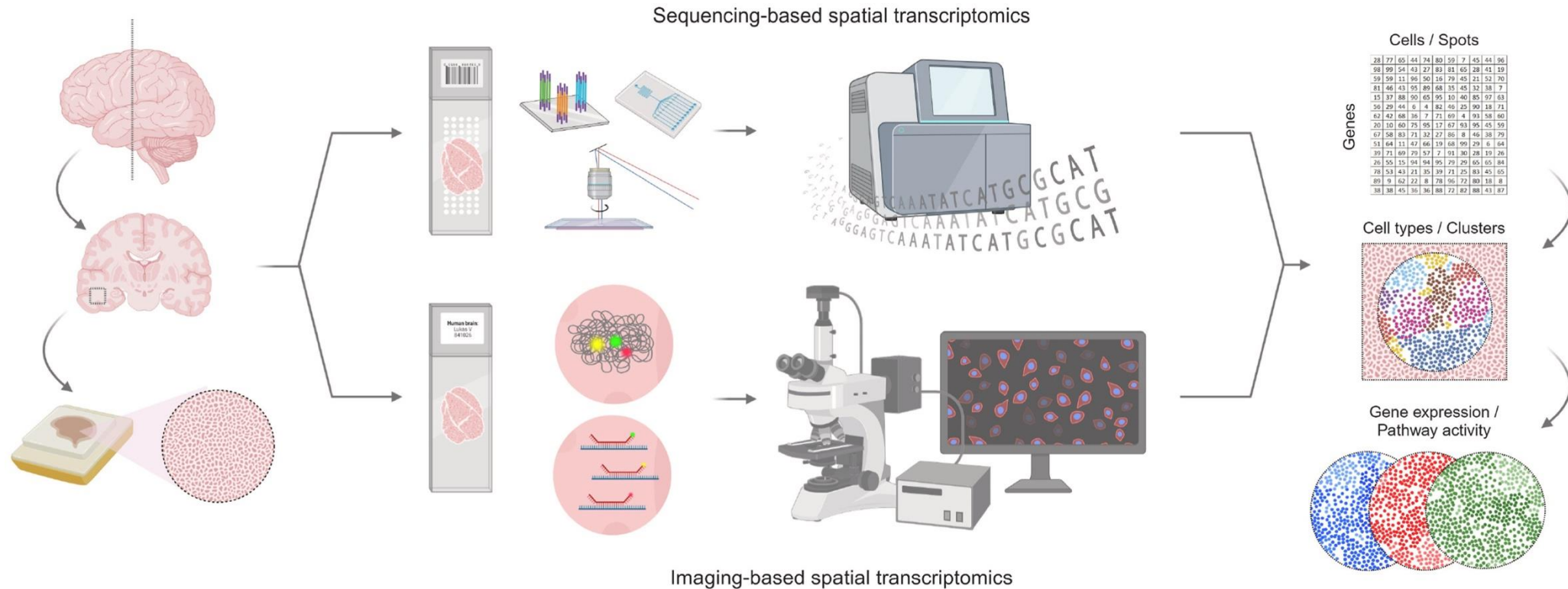


Introduction to spatial transcriptomics

History of spatial transcriptomics

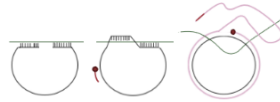


Modalities and technological platforms

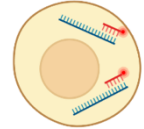


Imaging-based spatial transcriptomics

In-situ sequencing methods (ISS)



In-situ hybridization methods (ISH)



Examples:

ISS, STARmap, Barcode padlock probes, 10X Xenium

osmFISH, MERFISH, seqFISH(+)

Advantages:

- Allows 3D localisation of transcripts
- Independent of gene selection
- Better signal-to-noise ratio

- Targeted measurement
- Highly sensitive

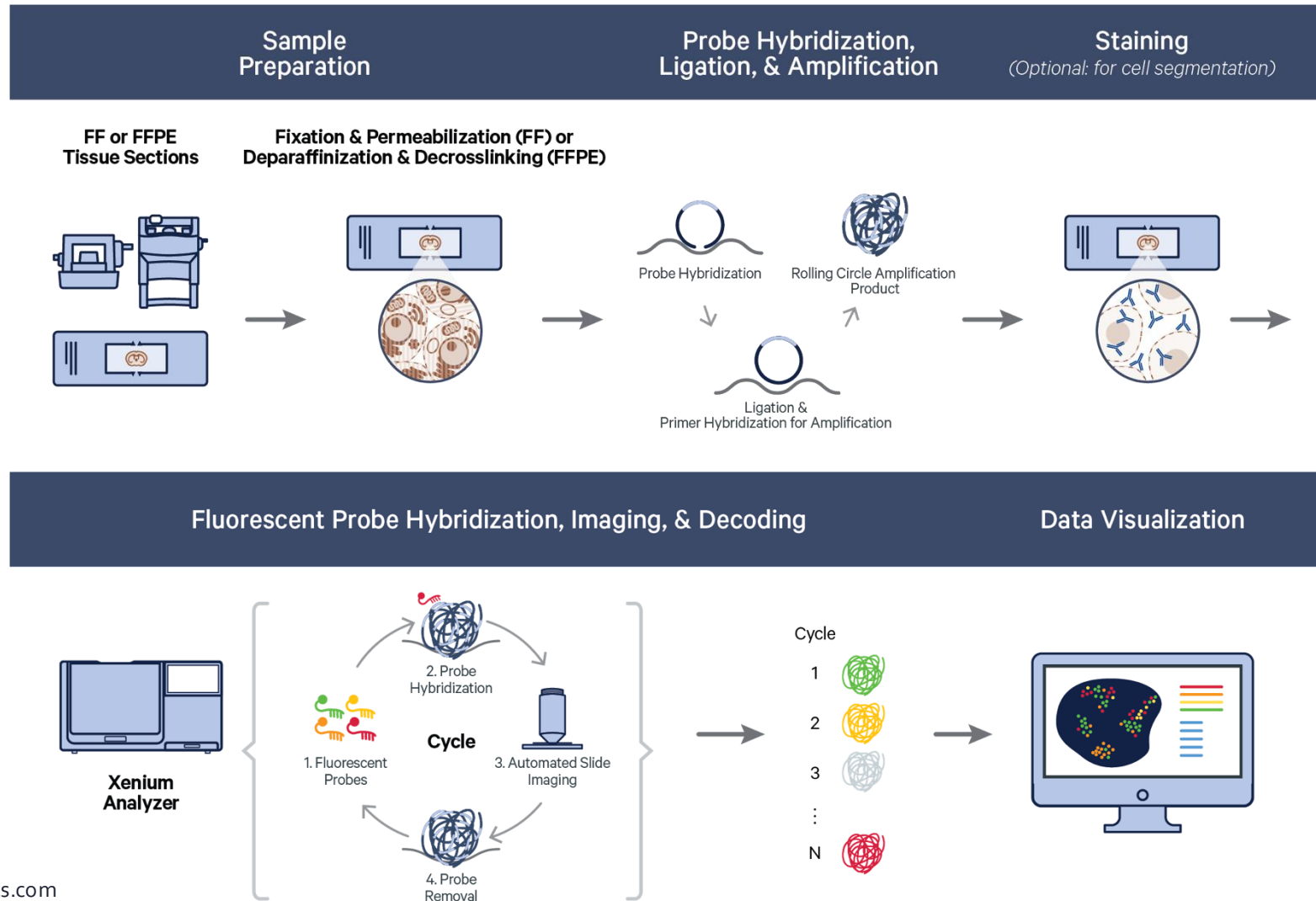
Limitations:

- Lower sensitivity

- Targeted measurement
- Labor expensive and expertise/equipment required

Xenium technology

Commercial in situ platform to use a padlock probe rolling circle amplification chemistry.



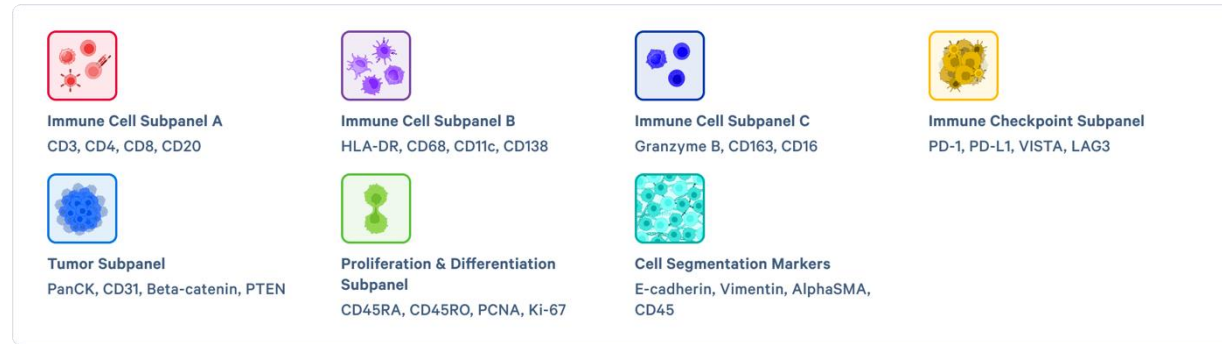
Visium technology

Xenium <500 genes

50-480 gene-panels
Fully custom panels (480 genes)
Add 100 custom genes

• **RNA V₁**

- 10.45mm x 22.45mm
- FFPE, FF, TMA's
- Human, Mouse, custom
- Same section: IF, H&E, Xenium protein panels



Xenium 5000 genes

5000 gene-panels
Add 100 custom genes

• **Prime**

- 10.45mm x 22.45mm
- FFPE, FF, TMA's
- Human, Mouse
- Same section: IF, H&E

Sequencing-based spatial transcriptomics

In-situ capturing (ISC)

Examples: 10X Visium (HD), slide-seq (vs), seq-Scope, Stereo-seq

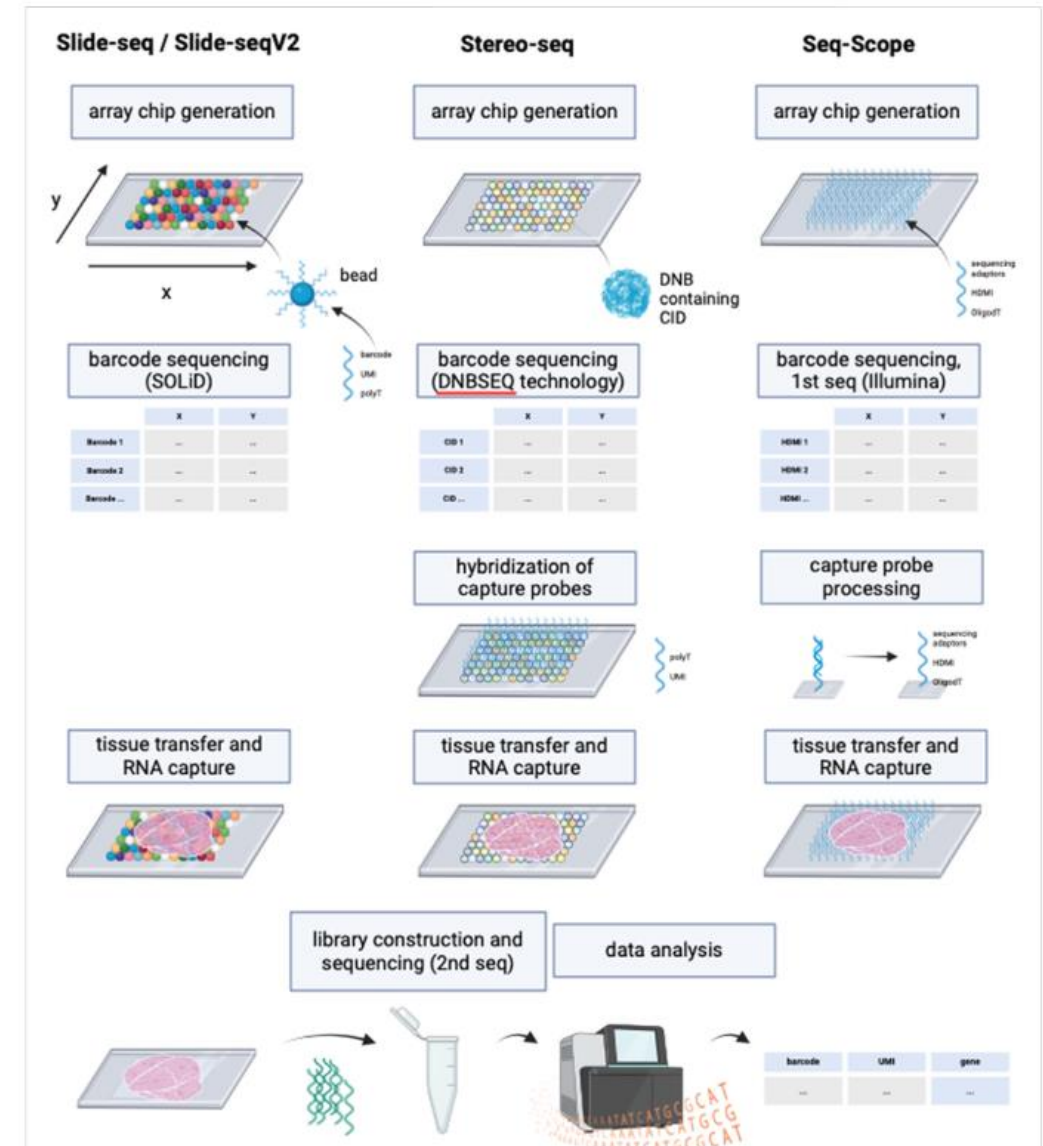
Major point of innovation is the **capture array** design.

Caputre array: clusters of oligonucleotides carrying specific sequence that encodes the spatial position. mRNA difuses from slice to the capture array. Ex-situ sequencing of the library.

Spatial measurement locations often referred to as “spots”, “beads”, or “bins”

Sequencing-based technologies

- Spatial transcriptomics (ST): capture oligos printed in glass slide
- Nanobeads: Oligos anchored to nano-beads (lower diameter)
Nanobeads are organised in a monolayer on the array
- Spatiotemporal enhanced resolution omics: DNA nanoballs locally amplified
Different chip sizes (including custom ones)
- Homogeneous oligonucleotide colonies



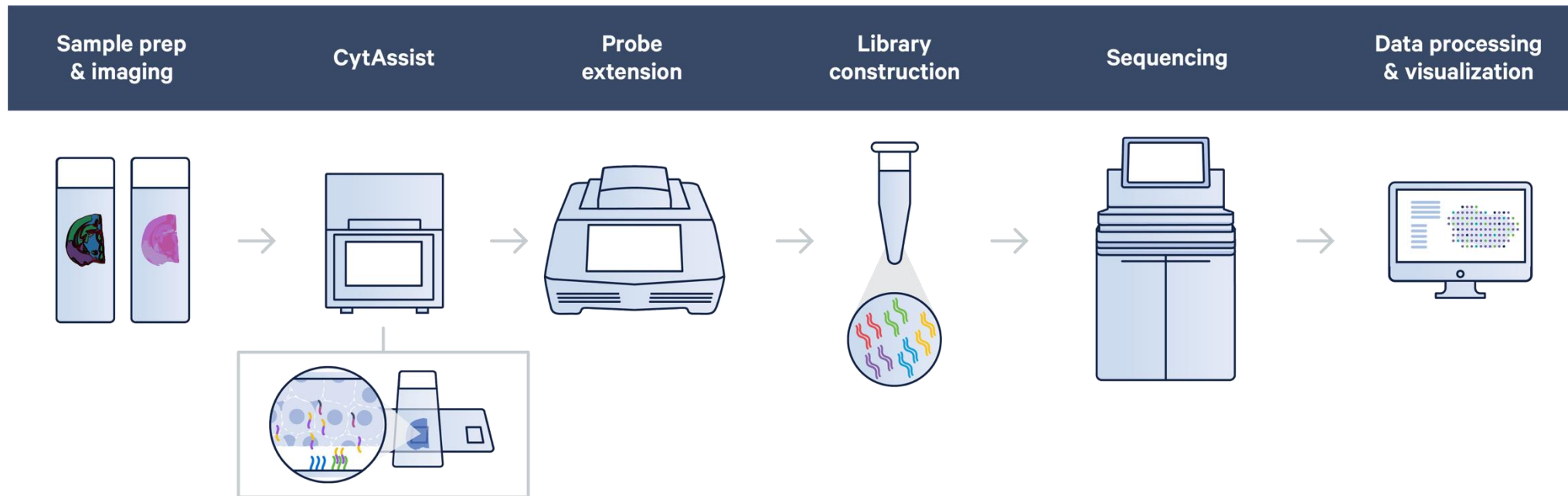
Sequencing-based technologies

methods	resolution (μm)	sensitivity/UMIs per 100 μm ² (tissue) ^a	capture area (mm ²)	detected target	accessibility ^b	ref
ST	100	31.88 (MOB) ^c	~40	polyA-tailed mRNAs	medium cost; commercialization as 10x Visium	(39)
10x Visium	55	508 (MOB)	~40	polyA-tailed mRNAs/target genes	high cost; commercially available	(77)
Ex-ST	~20	~496 (MOB)	~40	polyA-tailed mRNAs	high cost; tissue expansion technique and Visium slides	(40)
DBiT-Seq	10, 25, 50	~1320–4910 for mRNAs; 121.52 for proteins (ME) ^{c,d}	25 ^h	polyA-tailed mRNAs and ADTs ⁱ	medium cost; homemade microfluidic chip	(64)
Decoder-seq	15, 25, 50	4010 (MOB) ^e	25 ^h	polyA-tailed mRNAs	low cost; homemade microfluidic chip	(63)
xDBiT	50	200–800 (mouse organs)	116.64	polyA-tailed mRNAs	medium cost; homemade microfluidic chip	(62)
RRST	55	173.39 (MB)	~40	target genes	high cost; Visium slides and probe panel design	(44)
STRS	55	9–36 (mouse heart) ^f	~40	total RNAs	high cost; Visium slides	(45)
SHM-seq	100	~52 (mouse colon)	~40	host polyA-tailed mRNAs and microbiome rRNAs	high cost; Visium slides	(46)
SM-Omics	100	13.75 for mRNAs and 1.42 for proteins (MS) ^c	~40	polyA-tailed mRNAs and ADTs ⁱ	low cost; automated operating systems	(41)
spatial CITE-seq	25	78.88 for mRNAs; 35.4 for proteins (MS)	6.25	polyA-tailed mRNAs and ADTs ⁱ	high cost; ADTs and homemade microfluidic chip	(53)
spatial ATAC–RNA-seq	20, 50 ^g	189.36 (ME)	25 ^h	open chromatin and polyA-tailed mRNAs	NA; homemade microfluidic chip	(49)
spatial CUT&Tag–RNA-seq	20, 50 ^g	144.12 (P22 MB)	25 ^h	polyA-tailed mRNAs and histone modification	NA; homemade microfluidic chip	(49)
MISAR-seq	50	~450 (ME)	6.25	open chromatin and polyA-tailed mRNAs	NA; homemade microfluidic chip	(52)
Slide-seq	10	59 (E12.5 ME)	~7	polyA-tailed mRNAs	high cost; spatial decoding	(65)
Slide-seqV2	10	550 (E12.5 ME)	~7	polyA-tailed mRNAs	high cost; spatial decoding	(66)
HDST	2	12 (MOB)	13.68	polyA-tailed mRNAs	high cost; spatial decoding	(72)
Seq-Scope	<1	~1000 (ML)	0.8	polyA-tailed mRNAs	high cost; illumina flow cells	(68)
Pixel-seq	~1	977 (MOB)	315	polyA-tailed mRNAs	low cost; homemade polony gel	(69)
Stereo-seq	0.22	1450 (MOB)	up to 132 × 132	polyA-tailed mRNAs	high cost; has been commercialized	(67)
XYZeq	single cell	1.40 (ML ^c /tumor tissue)	~500	polyA-tailed mRNAs	NA; special facilities for tissue separation	(74)
sci-Space	single cell	41.23 (E14 ME)	324	polyA-tailed mRNAs	medium cost; sci-RNA-seq	(75)
Slide-tags	single cell	342.02 (E14 ME)	~7	polyA-tailed mRNAs, open chromatin, and T cell receptors	high cost; droplet-based snRNA-seq	(50)

Visium technology

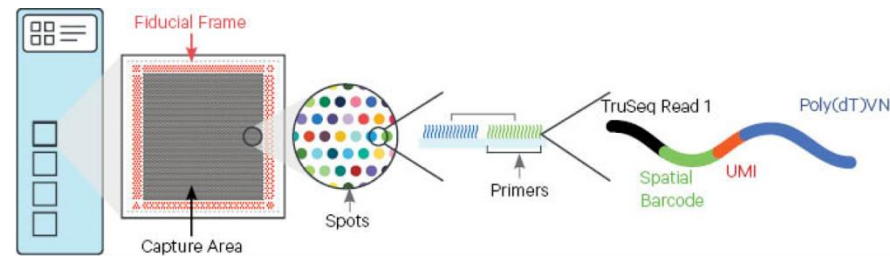
<https://www.youtube.com/watch?v=VwNk4d-oRJc>

10X spatial transcriptomics is similar to 10X scRNAseq technology, adding a spatially resolved barcode to the library



Visium technology

10x Visium



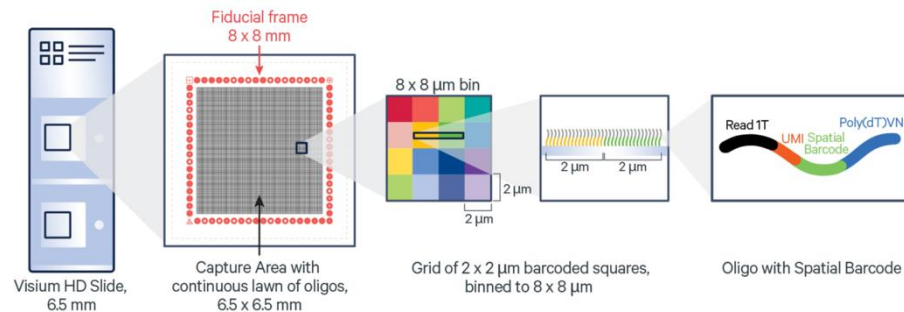
• Visium V1

- 3' poly(A) captured based
- Fresh frozen samples
- Diverse species
- 55μm barcoded spot (1-10 cells)

• Visium V2

- Probe based chemistry
- Fresh frozen, fixed frozen, FFPE samples
- Human and Mouse
- 55μm barcoded spot (1-10 cells)

10x Visium HD



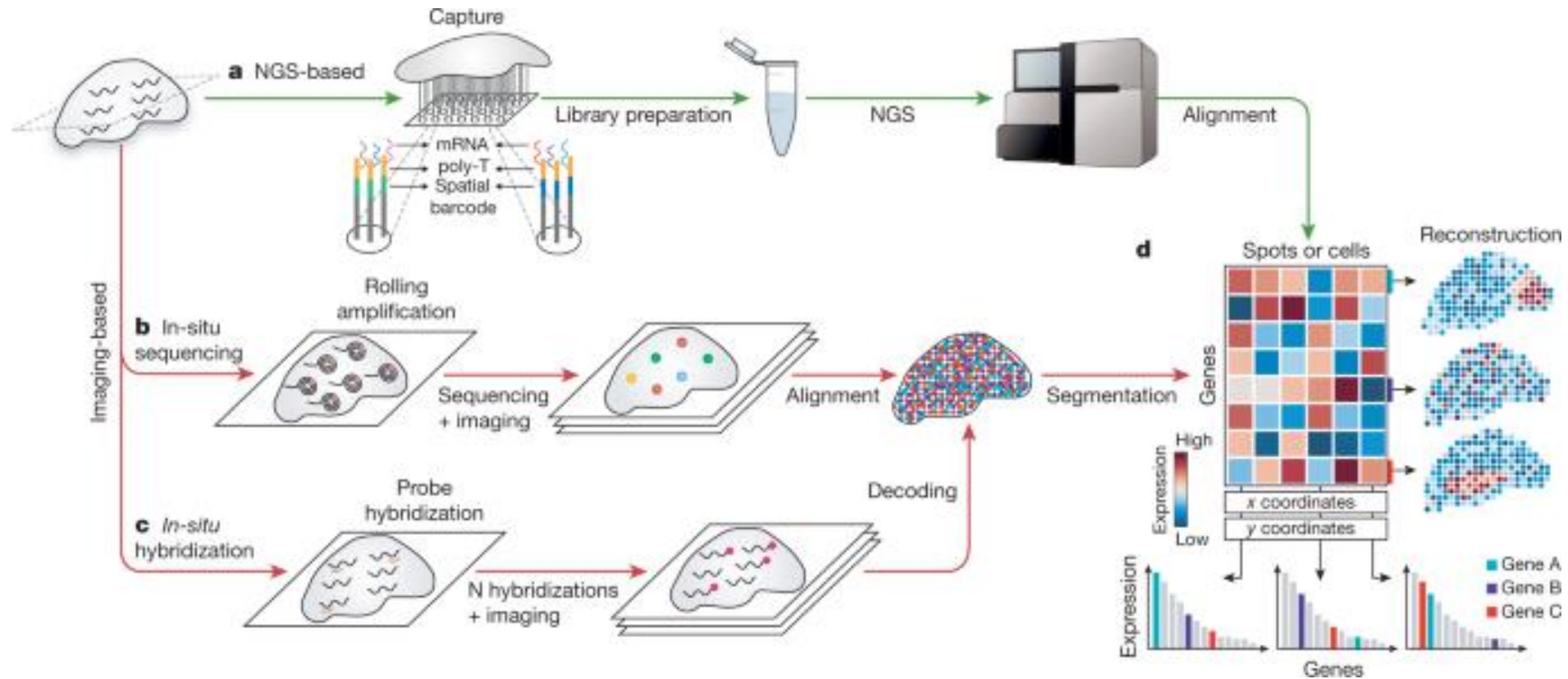
• Visium HD

- Probe based chemistry
- Fresh frozen, fixed frozen, FFPE samples
- Human and Mouse
- 2μm barcoded spot

• Visium HD 3'

- 3' poly(A) captured based
- Fresh frozen samples
- Diverse species
- 2μm barcoded spot

Imaging vs sequencing based technologies



Imaging vs sequencing based technologies



Imaging-based

- Higher spatial resolution
- Higher sensitivity

ISS
Xenium

- Lower genome coverage (targeted RNAs)
- Complex instrumentation needed
- Large amount of data generation

Sequencing-based

Visium HD

- Lower resolution
- Lower sensitivity (capturing efficiency)
- Less flexibility in sample preparation
- Higher genome coverage (untargeted)
- Simpler instrumentation and more scalable
- NGS (widely available)

Technologies keep evolving to overcome the limitations...

Choosing the right technology

Put your scientific question first, don't choose over methodology

- Gene throughput: unbiased or targeted?
- Sequence information: do you need actual sequence information?
- Sensitivity: interested in all genes? Known marker genes available? Highly expressed?
- Resolution: do you need single cell information?
- Feasibility: which access to tissues can you have? Are there reference datasets? Instrument/data analysis access? Cost?

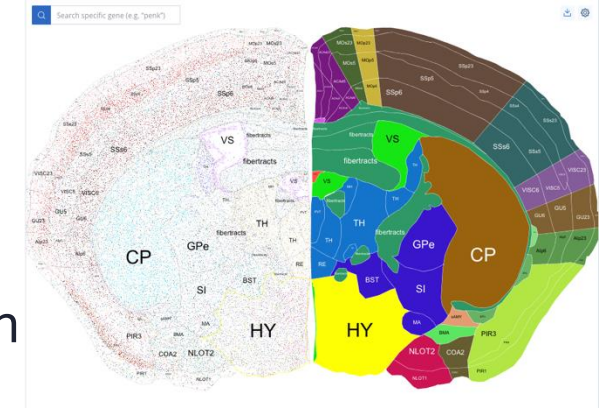
Very generalised!!!

Imaging-based: precise localisation of transcripts, annotate cell types within subregions, spatial interactions between cell types...

Sequencing-based: transcriptome-wide studies, new markers discovery, sequencing-read downstream analysis like velocity...

Applications

- Identification of cell types and tissue organisation: cell atlas, detection cell proportions in tissue structures
- Study of microenvironments and cell-cell interactions: exploration of many-to-many ligand receptor interactions, characterisation of tumour microenvironment
- Elucidate molecular and cellular gradients/patterns in complex tissues, providing insights into physiology, molecular biology and anatomy of organs
- Time resolved analysis: lineage tracing and cell fate inference



Mouse Spatial
Transcriptomics Atlas

Challenges and future directions

Limitations:

- Lacks multidimensional profiling
- Low capture sensitivity and RNA diffusion
- Compatibility with sample preparations
- Large amounts of data, problem to store, computational time and costs
- Scalability: Technology keeps scaling up to more resolution, more spots, etc.
Challenge for methods to scale up

Future directions:

- Holistic view with application of spatial multi-omics (epigenetics, proteomics, metabolomics...)
- Couple ST with long-read sequencing technologies
- 3D reconstruction of ST slices

